

2 Related Work

Multi-modal Large Language Models. Recent years have witnessed substantial advancements in Multi-modal Large Language Models (MLLMs). Early works like BLIP [41, 40] and Flamingo [5] introduce lightweight parameters between vision transformer [21] (ViT) and LLMs, laying the groundwork for multimodal perception. Subsequent efforts, such as LLaVA [45] and MiniGPT-4 [93], integrate instruction tuning, further enhancing the performance of MLLMs. Proprietary models like GPT-4o [33] and Gemini-Pro [64] have advanced MLLM performance on complex multimodal tasks, while open-source models such as Qwen-VL series [7, 70, 8] and InternVL series [15, 16, 24, 14, 94] achieve competitive results through optimized architectural design, dataset expansion and training paradigm improvements. Meanwhile, some related studies further advance the ability of large models by incorporating new modalities (e.g., audio [22, 19, 77], point clouds [27, 9], video [92, 12]) and by supporting more tasks (e.g., grounding [80, 71], computer usage [60, 6]). Notably, limited research attempts to enhance the reasoning capabilities of MLLMs. Some pioneering works, such as R1-Onevision [84], LMM-R1 [61], MM-EUREKA [58], R1-V [11], Visual-rft [50], Visualprm [72], OThink-MR1 [51], VLM-R1 [63], and Open-r1-Video [73] have explored the visual reasoning capabilities of MLLMs through Reinforcement Learning (RL), but they are still in the nascent stage.

Multimodal Benchmarks. With the development of MLLMs, multimodal benchmarks have also evolved significantly [43]. Early benchmarks primarily address visual perception tasks through simple tasks like visual question answering (VQA) [13, 44, 36, 78], image captioning [59, 20, 37] and referring expression comprehension [88, 55]. Subsequent works expand the capability coverage of benchmarks into more specialized domains: OCRBench [49], Chartqa [56] and DocVQA [57] assess textual content extraction; AgentBench [48] and ToolEyes [86] test tool usage capabilities; and egocentric perception benchmarks [54, 17] quantify first-person scene interpretation. Despite the progress, they ignore the evaluation of visual reasoning abilities [90, 89]. Recently, some benchmarks have made explorations in examining MLLMs’ visual reasoning abilities, but methodological deficiencies still cause limitations to assess the intrinsic visual reasoning capabilities [30, 4, 76]. InfIMM-Eval [29] test reasoning abilities around daily life, lacking deep-level reasoning scenarios. MMMU [89] and Emma [30] provide benchmarks demanding advanced reasoning abilities in fields such as chemistry and physics, but they ignore questions around the images’ fundamental visual components (e.g., shapes, elements). While mathematical benchmarks [69, 52, 31, 62, 91, 28] evaluate mathematical reasoning with geometric and diagram problems included, they focus on math capabilities but disregard logical analysis about the vision information. LogicVista [76] provides a multimodal logical reasoning benchmark, its visual questions lack analytical depth—dominated by single-hop, superficial queries in limited data scope. Unlike previous works, we introduce a challenging benchmark focused specifically on the domain of visual logical reasoning.

3 VisuLogic

In this section, we first describe the VisuLogic data-curation pipeline, which comprises three key stages: data collection, quality control, and the detailed taxonomy. We then report the benchmark’s construction statistics, including total size, answer-option distributions, and category-level proportions. Finally, we introduce a supplementary training dataset—consisting of questions analogous to those in VisuLogic—designed to bolster future research and facilitate community engagement.

3.1 Data Curation Pipeline

Data Collection. We construct the VisuLogic dataset by sourcing all questions from publicly available online resources in compliance with relevant licenses and regulations. As shown in Figure 4, our automated data processing pipeline comprises three stages: 1) **Fetching:** We employ Playwright¹ to systematically scrape raw web content, supplemented by custom parsing scripts that extract question-answer pairs. 2) **Cleaning:** We remove noise, irrelevant content, and extraneous HTML markup (e.g., `<div>`) to ensure the integrity of the textual data. 3) **Structuring:** We standardize the cleaned text and images by structuring all information (such as question text, metadata) in JSON Lines (JSONL) format.

¹<https://github.com/microsoft/playwright>

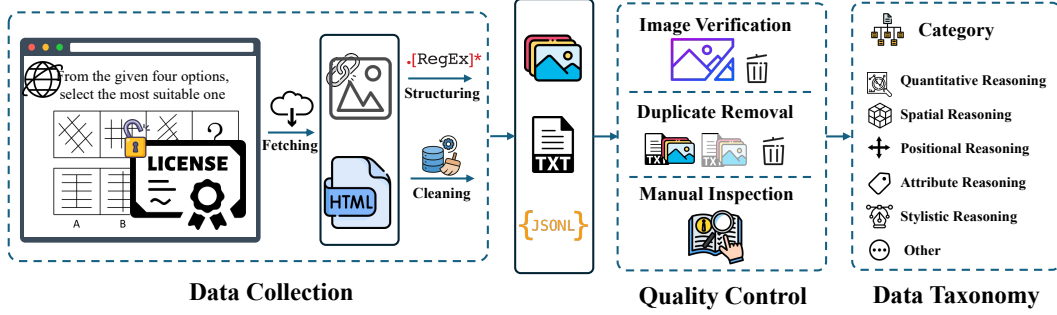


Figure 4: **Data curation pipeline of VisuLogic.** The pipeline includes Data Collection, Quality Control and Data Taxonomy.

Quality Control. To ensure the reliability of the benchmark dataset, we employ a three-stage data validation procedure: 1) **Image Verification:** Each image referenced in the questions is checked for existence and correct formatting; any item that fails to meet the criteria is removed following human review. 2) **Duplicate Removal:** We eliminate redundant entries at both the text and image levels by (i) detecting lexical overlap among text segments and (ii) applying perceptual hashing (pHash) to identify visually similar images. 3) **Manual Checking:** After automated filtering, we perform a thorough human-led review of every remaining entry to confirm its validity and ensure dataset reliability.

Data Taxonomy. We categorize all collected data into a taxonomy of six primary classes based on expert human annotation of the reasoning skills each question requires. Annotators first tag questions according to the targeted reasoning competency; these annotated tags are then analyzed and merged into five primary categories. A subsequent human review ensures that every question is accurately classified, with any ambiguous instances consolidated under the “Other” category. Specifically, we define each category as follows. **Quantitative Reasoning** focuses on changes in the number or count of graphical elements (for example, points, lines and angles) and on arithmetic relationships among shapes. **Spatial Reasoning** requires mentally reconstructing three-dimensional shapes from two-dimensional figures, folding or unfolding surfaces, and integrating three-dimensional structures. **Positional Reasoning** examines transformations such as translation, rotation and reflection of objects while preserving their fundamental elements. **Attribute Reasoning** involves intrinsic properties of shapes, including symmetry (axial or central), curvature and measures of openness or closedness. **Stylistic Reasoning** entails alterations in stylistic features such as overlay, subtraction and assessments of shape similarity or difference. **Other** encompasses questions that fall outside the preceding categories, including those involving letters, alphanumeric symbols or other specialized characters.

3.2 Dataset Statistics

Following data curation and validation, VisuLogic comprises 1,000 single-choice questions. Figure 1 (left) illustrates the category distribution: Quantitative Reasoning (35.3%), Spatial Reasoning (23.1%), Positional Reasoning (13.6%), Attribute Reasoning (8.2%), Stylistic Reasoning (9.0%), and Other (10.8%). Correct answer options are evenly balanced, with the proportions distributed as follows: A (23.1%), B (26.7%), C (25.2%), and D (25.0%).

3.3 Supplementary Training Dataset

To facilitate further investigation of visual reasoning, we provide an auxiliary training set of 4,296 question-answer pairs drawn from the same domains and subjected to identical validation procedures to prevent overlap with the benchmark. The training split mirrors the primary taxonomy, with category proportions of Quantitative Reasoning (30.7%), Spatial Reasoning (25.5%), Positional Reasoning (13.0%), Attribute Reasoning (8.8%), Stylistic Reasoning (9.9%), and Other (12.1%).

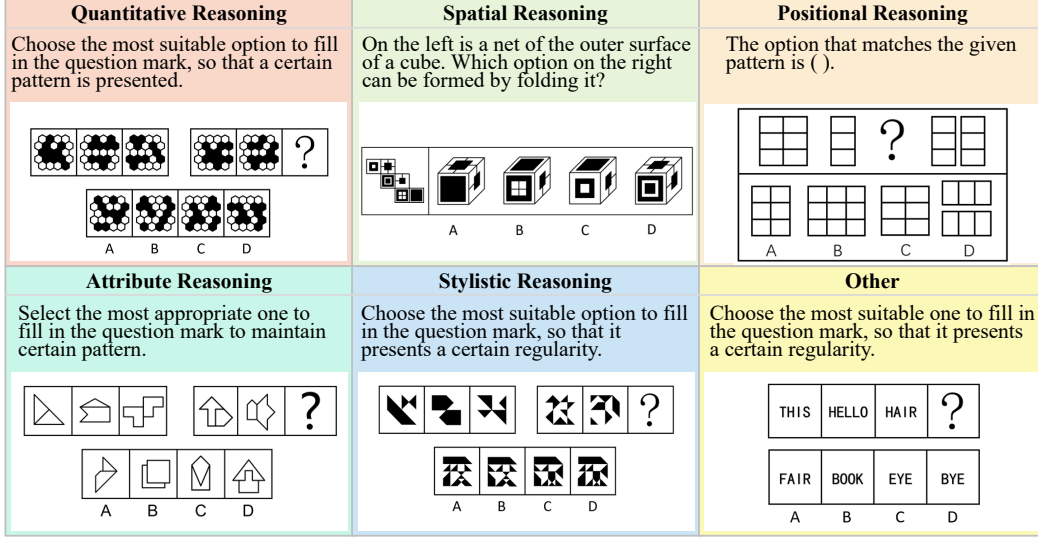


Figure 5: **Question examples of different categories in our VisuLogic Benchmark.** VisuLogic contains 6 categories of questions, which require models’ abilities in visual logic reasoning.

4 Experiments

In this section, we present a comprehensive evaluation of the VisuLogic benchmark. We first describe the experimental setup in Section 4.1, followed by overall performance results in Section 4.2. We then analyze systematic errors in Section 4.3 and provide qualitative insights in Section 4.4.

4.1 Experiment Setup

References Performance. To fully investigate models’ performance, we establish two reference points: 1) **Human Performance:** We invited 100 graduate students majoring in science and engineering to solve 10 randomly sampled VisuLogic questions each, allowing 2–5 minutes per question. The aggregate accuracy over all participants constitutes the human benchmark. 2) **Random Selection:** We simulate random guessing by sampling answers uniformly over 10 independent runs and report the average accuracy as the random baseline.

Evaluated Models. We evaluate a total of 28 models on VisuLogic, comprising 8 large language models (LLMs) and 20 multimodal large language models (MLLMs). For open-source LLMs, we test *Deepseek-R1* [18], *Qwen2.5-72B-Instruct* [83] and *Qwen-QwQ* [67], and for close-source LLMs we evaluate *GPT-4* [1], *o3-mini*, *Gemini-2.0-Flash-Thinking* [64], *Claude-3.7-Sonnet* and *Doubao-1.5-Pro-32k*. Open-source MLLMs include *Qwen2.5-VL-7B-Instruct* [8], *Qwen2.5-VL-72B-Instruct* [8], *QvQ-72B-Preview* [66], *InternVL2.5-38B* [16], *InternVL2.5-78B* [16], *InternVL3-38B* [94], *InternVL3-78B* [94], *LLaVA-v1.5-7B* [46], *LLaVA-OneVision-7B (SI)* [39], *ShareGPT4V* [10], *MiniCPM-o-2.6* [85], *GLM-4v-9B* [25], *Ovis2-8B* [53] and *mPLUG-Owl3-7B* [87], while close-source MLLMs consist of *GPT-4o* [33], *GPT-4o-mini*, *Kimi-latest* [65], *Doubao-1.5-Vision-Pro-32k*, *Gemini-2.0-Pro* [64] and *Claude-3.7-Sonnet*. We further include two reinforcement-learning baselines built on *Qwen2.5-VL-7B-Instruct* [8] and *InternVL2.5-38B* [16], respectively, trained via our rule-based RL procedure on our supplementary training dataset. Fully supervised fine-tuning (SFT) experiments on the same datasets serve as controls to isolate the effect of RL optimization. All model hyperparameters, training regimes, and implementation details are provided in the Appendix.

LLM Evaluation Protocol. For language-only models, we generate an auxiliary image description using GPT-4o and prepend it to the question. Specifically, each question is formatted as “Following is a detailed caption describing an image: [IMAGE DESCRIPTION]. Based on the provided description, select the best answer from the four options:”. This combined prompt is fed directly into the target LLMs for inference.